Here is the detailed report for **Question 1**, specifically tailored based on the code and results found in the notebook UntitledANSHU.ipynb.

Report: Rule-Based Sentiment Analysis (Question 1)

1. Task Objective

The primary goal of this task is to perform sentiment analysis on financial news headlines using two distinct rule-based approaches. We aim to evaluate whether a specialized financial dictionary (Loughran-McDonald) performs better than a general-purpose sentiment tool (VADER) when dealing with complex financial terminology.

2. Text Preprocessing Pipeline

To prepare the raw headlines for analysis, we implemented a structured preprocessing pipeline in the notebook:

- **Cleaning:** Used regular expressions (re) to remove punctuation, digits, and special characters. All text was converted to lowercase.
- **Tokenization:** Sentences were split into individual words (tokens) to facilitate word-by-word matching.
- **Stopword Removal:** Common English words (e.g., "the", "and") were removed using the NLTK library to focus only on words that carry sentiment.
- **Lemmatization:** We utilized the **WordNetLemmatizer**.
    - **Justification:** We chose lemmatization over stemming because it reduces words to their meaningful base form (lemma). In financial text, it is crucial that words like "banking" and "banks" are reduced to "bank" to ensure they are correctly matched against the sentiment lexicon.

3. Methodology

A. Loughran-McDonald (LM) Dictionary

We utilized the specialized LM financial lexicon, specifically targeting 'Positive' and 'Negative' word categories.

- **Implementation:** The code uses **Python Sets** for the positive and negative word lists.
- **Optimization:** Using Sets is a critical optimization. It allows for **O(1) lookup time** on average, making the matching process significantly faster than using standard lists, which would require O(N) time.
- **Logic:** A headline is labeled **POSITIVE** if the count of positive words is greater than negative words, and **NEGATIVE** if vice versa. If counts are equal, it is labeled **NEUTRAL**.

B. VADER Sentiment Analysis

VADER (Valence Aware Dictionary and sEntiment Reasoner) was used as the general-purpose benchmark.

- **Logic:** It calculates a **Compound Score** (normalized between -1 and +1).
- **Thresholds:** * **Positive:** Score > 0.05
    - **Negative:** Score < -0.05
    - **Neutral:** Score between -0.05 and 0.05.

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4. Outcomes and Results

Based on the code execution in UntitledANSHU.ipynb, here are the performance metrics generated:

Accuracy Metrics Comparison Table

Method	Correct Positive %	Correct Negative %	Overall Accuracy %
Loughran-McDonald	46.53%	**65.92%**	**58.46%**
VADER	**82.27%**	39.01%	51.55%

Code-Based Observations:

- **LM Performance:** The Loughran-McDonald method showed a much higher accuracy in identifying **Negative** sentiments (65.92%) compared to VADER. This is because financial "negativity" (e.g., "liability," "litigation") is often missed by general tools.
- **VADER Performance:** VADER was very aggressive in labeling things as **Positive** (82.27% accuracy in that class), but it failed significantly in identifying Negative financial news (only 39.01% accuracy).

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5. Conclusion

The specialized **Loughran-McDonald** lexicon is more reliable for overall financial sentiment analysis (58.46% accuracy) compared to VADER (51.55%). The results confirm that general-purpose tools like VADER struggle with domain-specific jargon where words that are "neutral" in daily life have a "negative" impact on stock prices.